STUDENT PERFORMANCE

& RETENTION ANALYTICS

Group F | Business Intelligence & Data Visualization

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Project Overview

Objective

To analyze student performance in online learning environments using the Open University Learning Analytics Dataset (OULAD), identifying patterns influencing student success, failure, and withdrawal.

~29,000

Total Students

Filtered cohort

36%

Success Rate

Course completion

33%

Withdrawal Rate

Dropped out

Tools Used:

- Microsoft Power BI
- Power Query (ETL)
- DAX Measures
- OULAD Dataset
- GitHub Repository

Data Sources

Open University Learning Analytics Dataset (OULAD) - Kaggle / OU Open Data

studentInfo — Demographics

- Age band (split into min/max), gender, region
- Highest education, disability status
- Number of credits studied, final result

studentVle — Interactions

- Daily click activity on VLE resources
- Activity type (forum, resource, quiz…)
- Days since module start (engagement timing)

studentAssessment — Scores

- Assessment scores (0–100)
- Submission dates → days to deadline
- Assessment type: TMA, CMA, Exam

assessments / courses / vle

- Assessment weights and types
- Module-presentation details
- VLE resource catalogue

Data Preparation & Cleaning

Power Query Transformations (10+)

- Replaced nulls with 0 in weight/score columns
- Changed data types (dates, numbers, text)
- Removed irrelevant columns (is_banked)
- Conditional column: performance_category (High/Avg/Low)
- Split imd_band → imd_band_lower + imd_band_upper
- Custom column: imd_band_avg (average deprivation score)
- Conditional column: imd_category (High/Mod/Low Deprivation)
- Split age_band → age_min + age_max (numeric)
- Conditional column: age_category (Young/Middle/Older)
- Renamed date columns → days_since_start, days_to_submission
- Conditional column: activity_period (Before Start / Early / Mid / Late)
- Conditional column: submission_period (Early / On-time / Late / Very Late)

Calculated Metrics (DAX)

Completion Rate

% students with final_result = 'Pass'

Dropout Rate

% students who Withdrew from course

Pass Rate

% assessments with score ≥ 40

Average Score

Mean assessment score across all students

At-Risk Students

Score < 40 AND VLE clicks < 100

Avg Clicks / Student

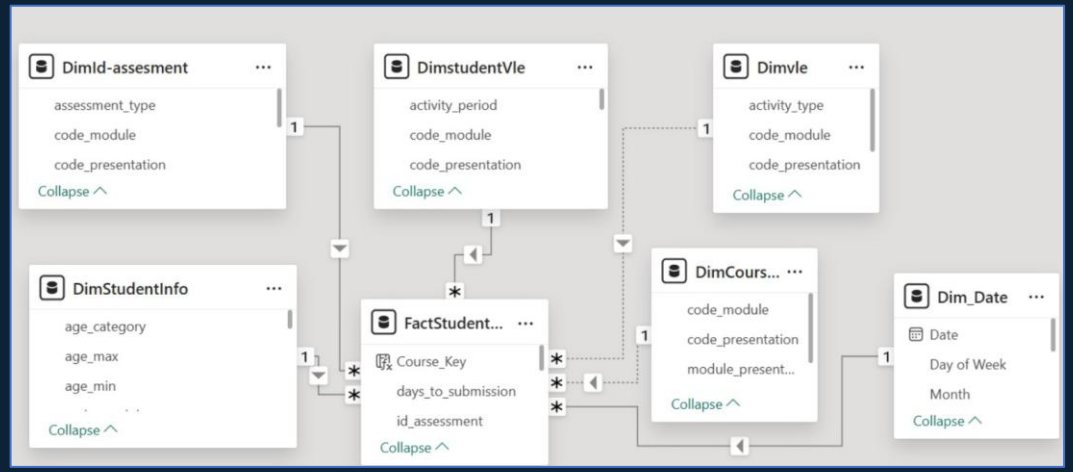
Total VLE clicks ÷ distinct students

Course_Key

Surrogate key linking assessments to courses

Model View

Star Schema



Fact Table

FactStudent — central transactions: assessments, submissions, scores

Dimensions

6 dimension tables: Students, Assessments, VLE, Courses, Date, Activity

Relationships

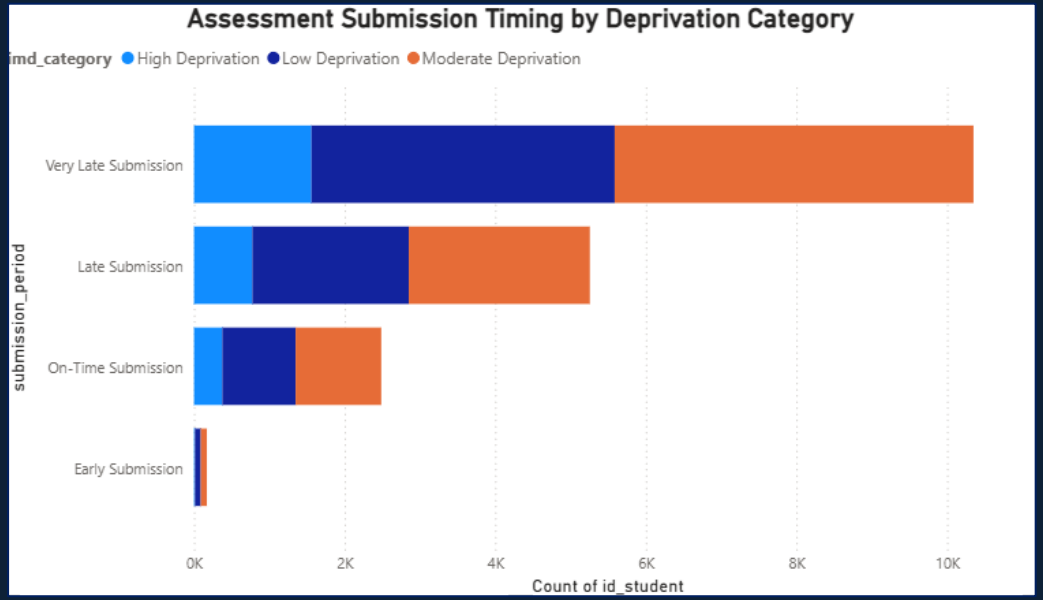
Solid lines = active 1:Many | Dotted = inactive (conflict avoidance)

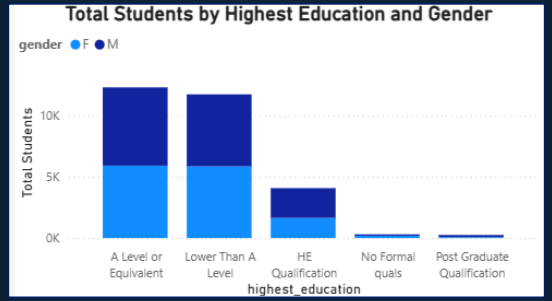
Date Dim

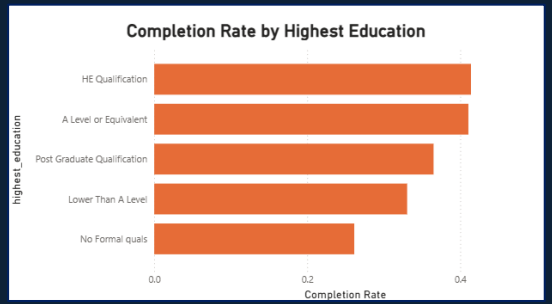
Dedicated Date table for all time-intelligence DAX measures

Dashboard

0.96	0.39	0.30	14K
Pass Rate	Completion Rate	Dropout Rate	At Risk Students





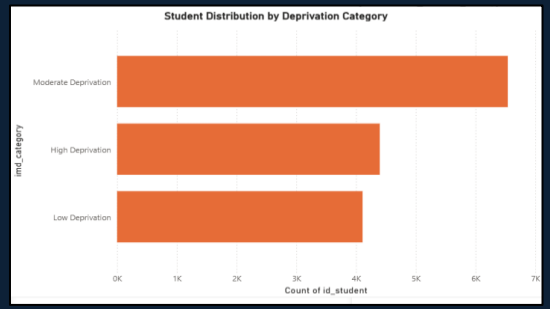


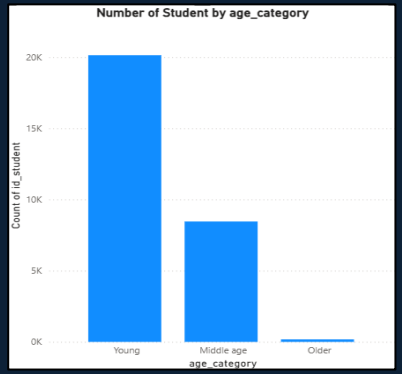
Higher deprivation = later submissions, lower engagement

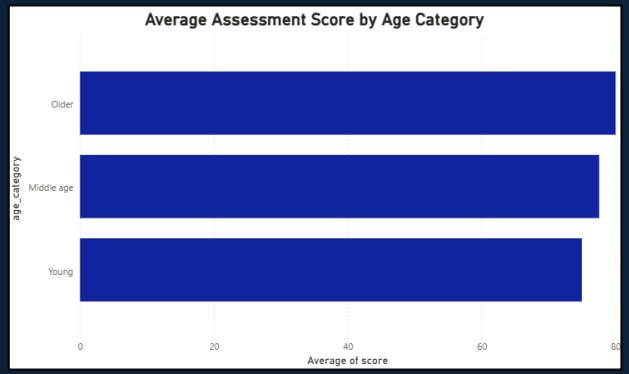
HE-qualified students: ~50% completion vs ~20% for no qualifications

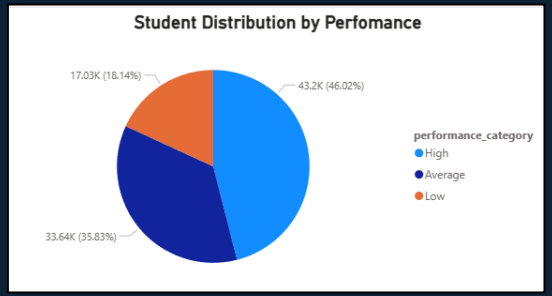
Low early engagement + late submission = highest dropout signal

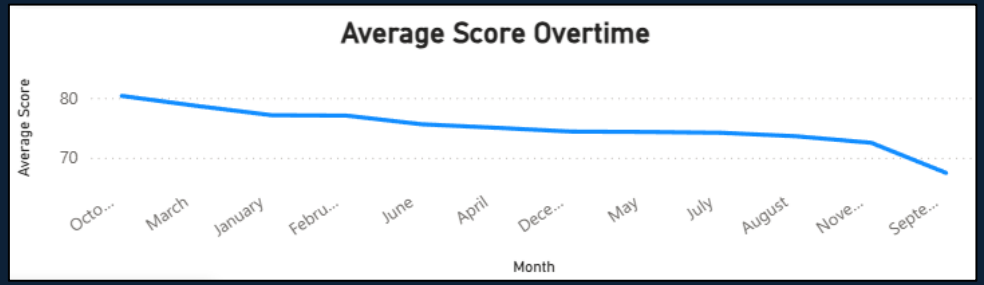
Dashboard Visualizations











Young learners dominate (70%), but older students outperform

~10,000 students submit very late — the single largest group

Students interacting before module start: 26% more likely to succeed

Key Insights

Demographics

- Older students consistently outperform younger peers, maturity and self-regulation are key factors
- Students with higher prior education show significantly higher completion rates
- Students from higher deprivation backgrounds are more likely to submit assessments late.
- Socio-economic disadvantage is associated with lower engagement consistency and performance.

Academic Load

- Students taking excessive credit loads score lower on average, cognitive overload is a risk
- Balanced credit workloads strongly correlate with better performance outcomes
- Average assessment scores decline gradually across the module, indicating possible learning fatigue or increasing assessment difficulty.

Key Insights

Engagement

- Early and consistent VLE engagement is the strongest predictor of success
- Students interacting before module start perform significantly better than late starters
- Engagement declines as the module progresses, aligning with decreasing average scores over time.

At-Risk Indicators

- Late submissions + low early engagement = high dropout probability
- Limited prior education combined with low VLE clicks flags highest-risk students

Recommendations

01

Early Warning Systems

Flag students with low early VLE engagement (<100 clicks in first 30 days) for proactive outreach before they fall behind.

02

Targeted Support Programs

Provide supplementary academic support to students with lower prior education qualifications, especially in early module weeks.

03

Engagement Incentives

Encourage early and consistent engagement through structured pre-module orientation and gamification of VLE resources.

04

Workload Management

Advise students against taking excessive credit loads. Use predictive analytics to recommend appropriate credit loads per student profile.

05

Continuous Monitoring

Deploy real-time dashboards enabling tutors to track at-risk indicators and intervene with personalized feedback mid-module.

06

Predictive AI Integration

Extend the dashboard with machine learning models to predict likelihood of withdrawal at the start of each module.

Challenges & Limitations

Data Quality Issues

- Null values in weight and score columns required imputation
- Age bands and IMD bands stored as text ranges, needed splitting and conversion
- Date columns stored as integers (days from start), not actual calendar dates

Modeling Constraints

- VLE click data measures quantity, not quality of engagement
- External factors (employment, personal life) not captured in dataset
- Correlational findings; cannot establish direct causation

Analytical Assumptions

- Pass threshold assumed at score ≥ 40
- At-risk defined as score < 40 AND VLE clicks < 100, thresholds are approximate
- Filtered cohort may not fully represent entire student population

CONCLUSION

- **Only 36% of students complete their courses,** highlighting a critical institutional performance gap.
- **Engagement, demographics, and academic load** are the key interconnected drivers of student outcomes.
- **Early intervention using VLE data** can identify at-risk students before they withdraw.
- **Data-driven decision-making** is essential for improving online learning completion rates.

Next Steps: *Predictive analytics modeling • Real-time student monitoring • Personalized AI-powered feedback*